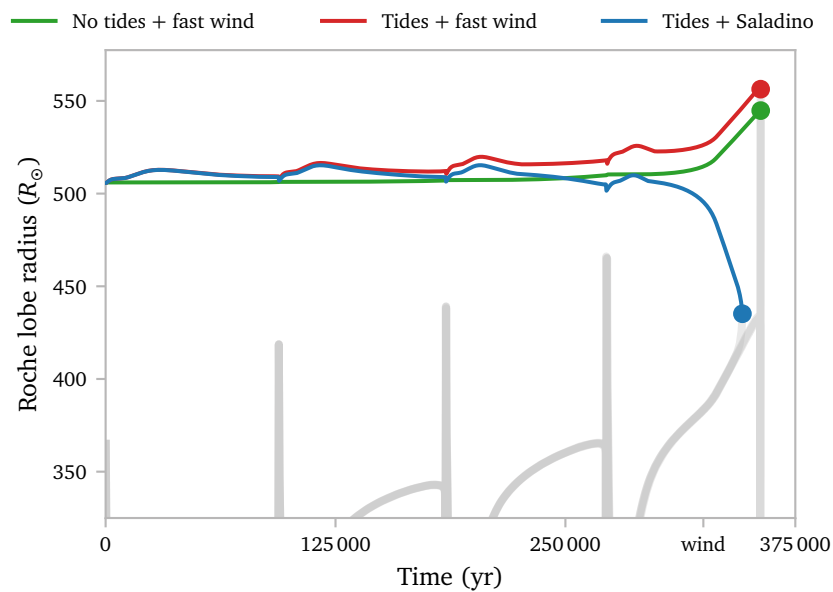


Notes Week 10

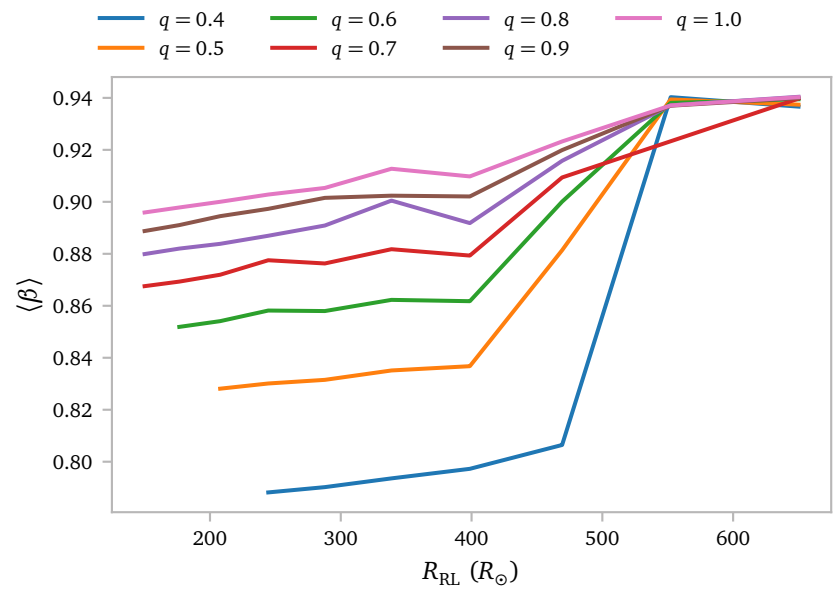
a. Stellar Evolution Meeting



b. Radiation Corrected Transfer Rate

As described by [Temmink et al. \(2025\)](#), the MESA setting `use_radiation_corrected_transfer_rate` is indeed on by default¹. From now on, I disable this in the binary inlist.

I check the overall mass transfer efficiency for the models from the last grid.



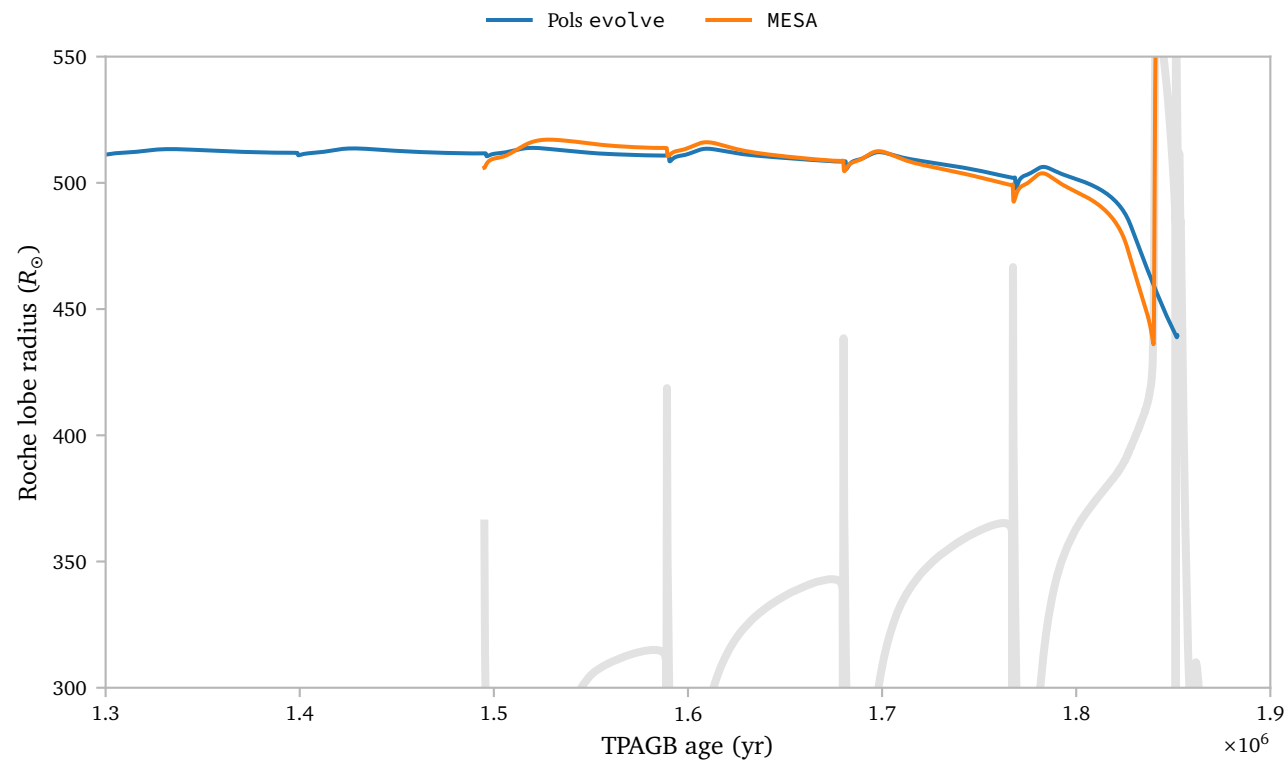
¹ When I looked at this before, I checked the MESA wiki for the latest stable release. Here, it mentioned that this option is *disabled* by default, but this has only changed with the latest release.

There is quite a strong dependence on the mass ratio. The efficiency is determined by

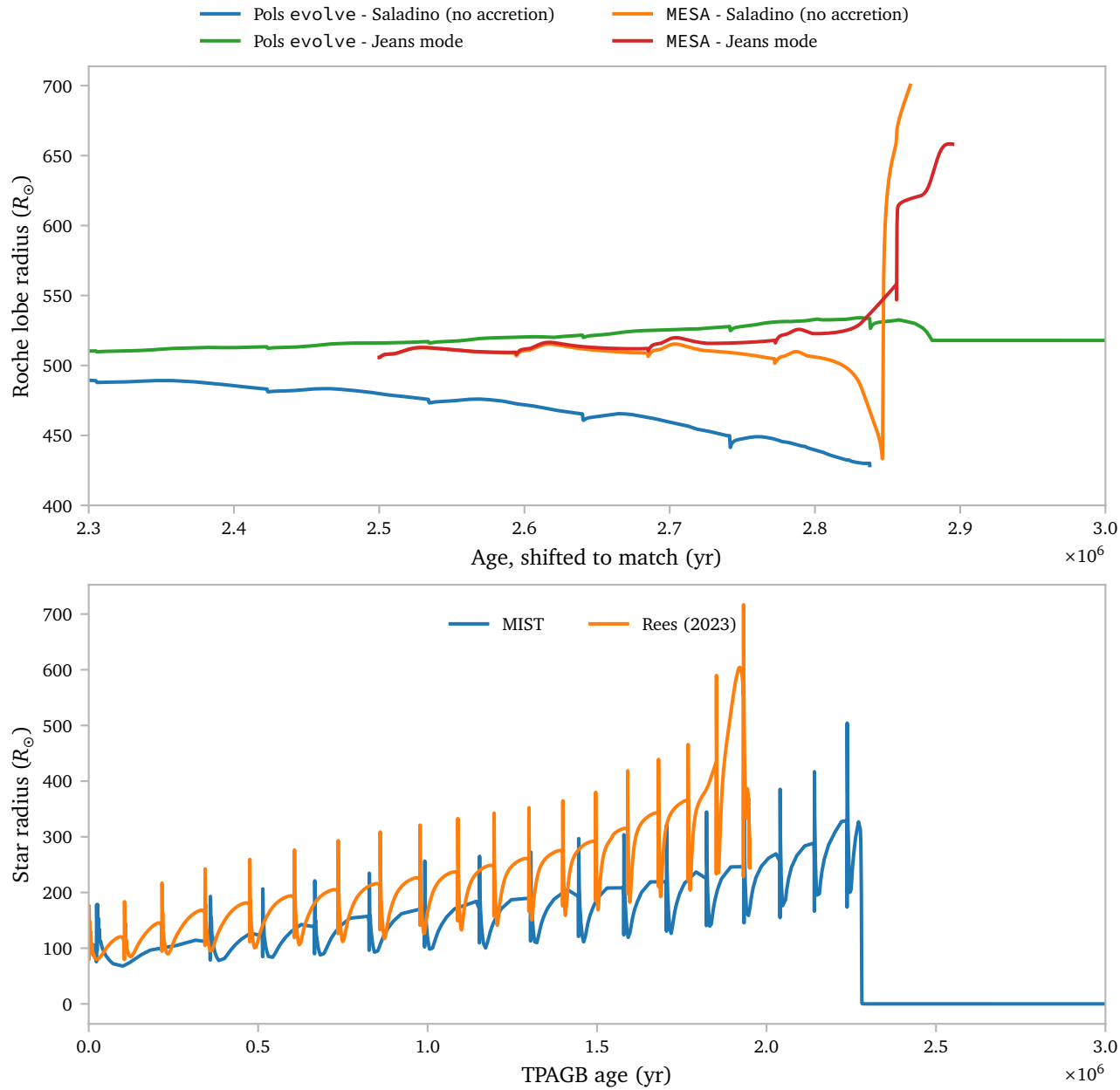
$$1 - \eta = \sqrt{1 - \left(\frac{M_{BH}}{3M_{BH,0}} \right)^2}.$$

For lower q , this will generally result in lower efficiencies. If mass transfer is nearly completely avoided, the initial and final black hole mass will roughly be equal and the whole expression reduces to $1 - \eta = 2\sqrt{2}/3 \approx 0.943$, which we can see for large Roche lobe radii where mass transfer is avoided.

c. evolve.py-script with Rees Single Star TPAGB Models



Below I show the comparison figure from last week as well. It is clear that,



when using the same star as input, the results are a lot more consistent. Obviously there are still some differences, especially during the initialization of the MESA model, which starts synchronized². I imagine the other differences can be explained by the way the two models compute the convective regions and other relevant properties.

² Which is generally not a bad assumption, but if a model starts during a thermal pulse, which most do, the star would usually be out of tidal synchronization.

d. Wind accretion

As mentioned last week, we need to take accretion of the secondary into account when we use the wind prescription by [Saladino et al. \(2019\)](#). In this prescription, β describes how much of the wind mass lost from the donor gets accreted by the accretor. Since we already have a nice way to compute this β , we can simply use the same methods to find β for the change in mass, and hook this method into the routine `other_binary_wind_transfer`. Below, I write the full routine:

```

subroutine wind_accretion(binary_id, s_i, ierr)
  integer, intent(in) :: binary_id, s_i ! s_i is index of the wind mass losing star
  integer, intent(out) :: ierr

  type(binary_info), pointer :: b
  type(star_info), pointer :: s

  real(dp) :: q, vw_over_vorb

  call binary_ptr(binary_id, b, ierr)
  if (ierr /= 0) then
    write (*, *) 'failed in binary_ptr'
    return
  end if

  ! --- compute q == m_accretor / m_donor and v
  q = b%m(b%a_i)/b%m(b%d_i)
  vw_over_vorb = compute_vwind(b%s_donor)/compute_vorbit(b%m(1) + b%m(2), b%separation)

  ! --- get beta from Saladino
  beta = compute_beta_Sal(q, vw_over_vorb)/sqrt(1 - pow2(b%eccentricity))
  b%wind_xfer_fraction(s_i) = beta

end subroutine wind_accretion

```

It is important to tell MESA to actually compute wind accretion³. Although I linked my custom subroutine to `other_binary_wind_transfer` using

³ Which I forgot to do initially.

```
b% other_binary_wind_transfer => wind_accretion
```

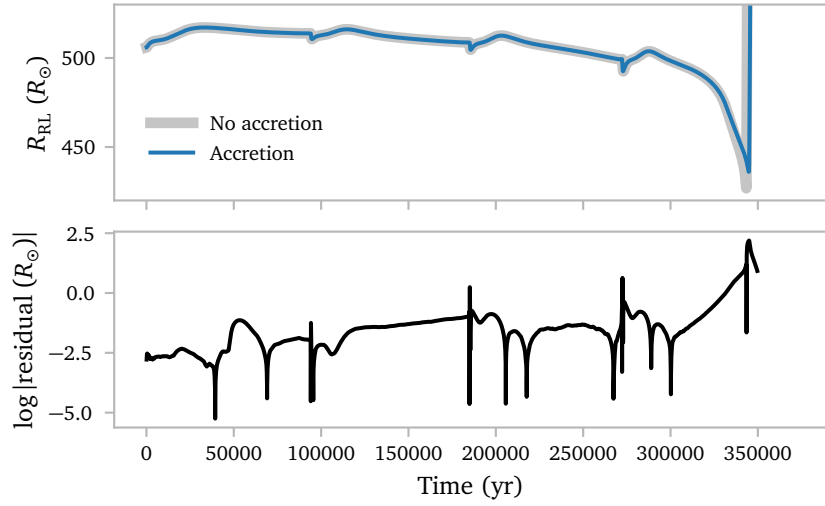
and set this in the inlist of the binary as well using

```
use_other_binary_wind_transfer = .true.
```

MESA still does not automatically actually use the wind mass transfer routine. For this, one *must* enable the option

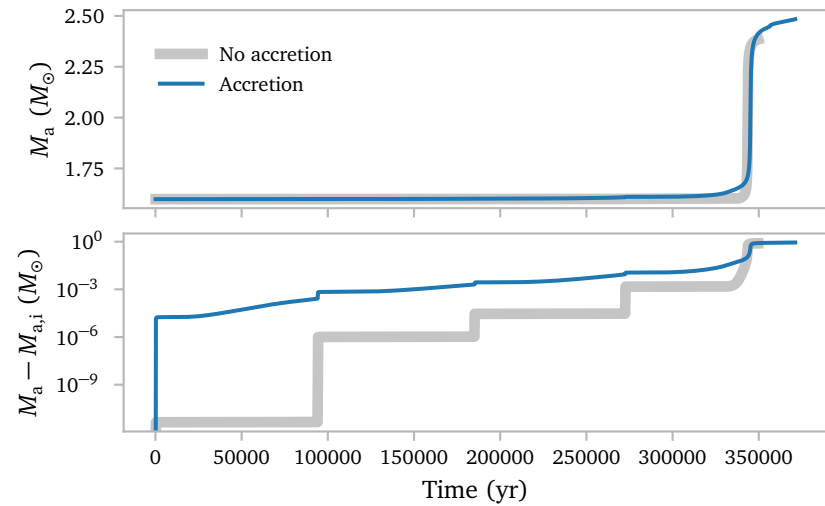
```
do_wind_mass_transfer_1 = .true.
```

Below, I show the results of a run with, and without wind mass accretion.



There is very little difference between the Roche Lobe radii of both models. It only becomes slightly noticeable when the wind starts to ramp up just before RLOF.

More interestingly maybe, let's look at the mass of the secondary.



The accretor clearly accretes mass expelled as wind from the donor!

e. Validating Tides + Wind Prescription

e.1 Tides

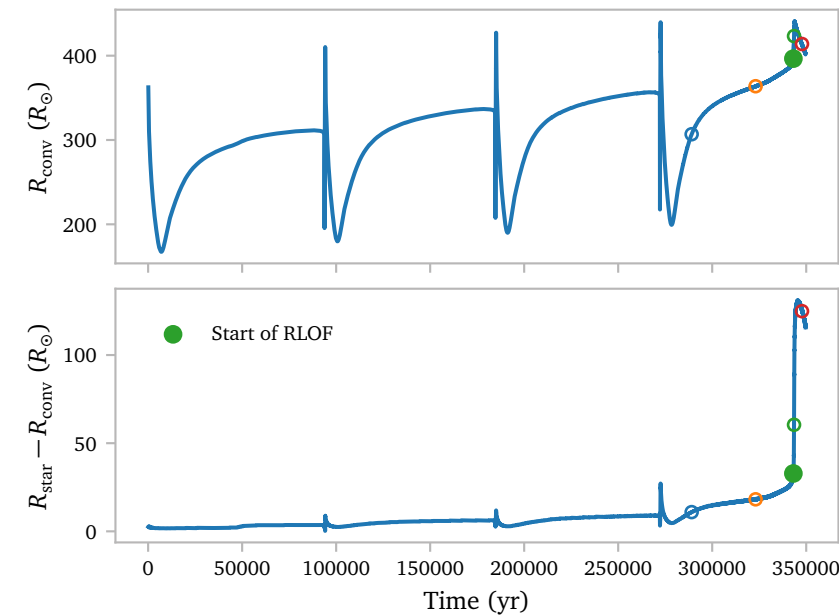
In order to ensure the custom tides and wind prescriptions are working as expected, I try to thoroughly check every component of the implementation. I start with the computation of the ratio k/T^4 . I use the prescription by Preece et al. (2022) which uses a power law of the form

$$\left(\frac{k}{T}\right)_c = \left(\frac{R_{\text{conv}}}{R}\right)^a \left(\frac{M_{\text{conv}}}{M}\right)^b \frac{c}{t_{\text{conv}}}.$$

Clearly, this ratio depends on the mass and radius of the convective envelope, and the so called convective turnover time t_{conv} . Preece et al. (2022) defines this as

$$t_{\text{conv}} = 0.4311 \left[\frac{3M_{\text{conv}}R_{\text{conv}}^2}{L} \right]^{1/3} \text{ yr.}$$

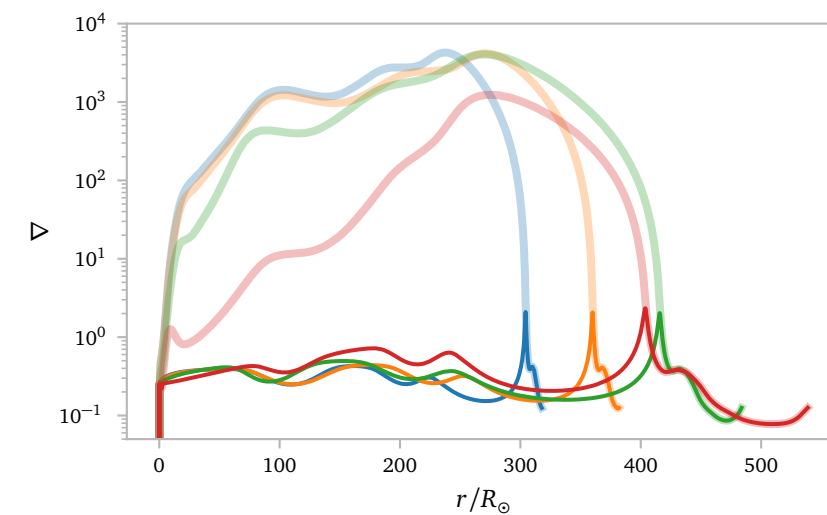
I start by looking at R_{conv} , M_{conv} and t_{conv} for a reference run.



⁴ Which is used in the tidal prescription by Hut (1981) and encodes the structure of the star.

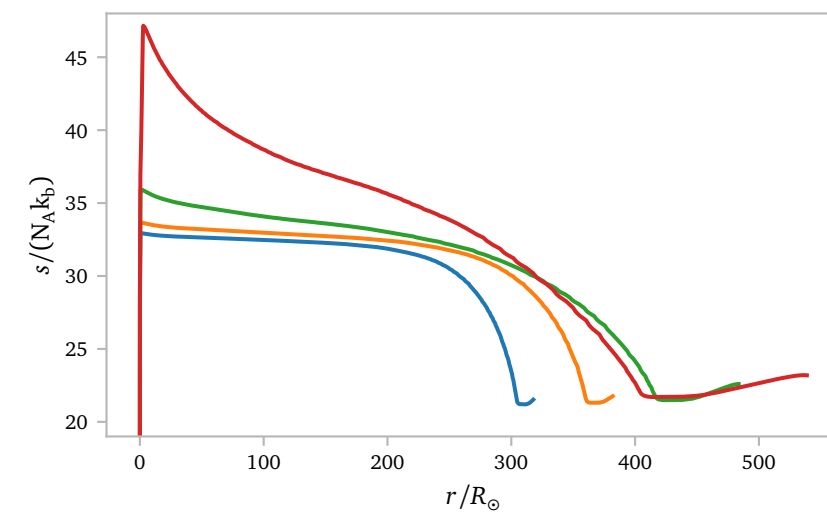
I think it looks good, the radius of the convective envelope is roughly the radius of the star, which is what I would expect. I do however find it a bit weird that the convective envelope seems to become much smaller than the stellar radius during / after RLOF I want to look into this before I continue.

I therefore investigate some of the profiles of the donor. I label the points of the profiles in the graphs with the open dots. The colors correspond with the lines of the plot below.



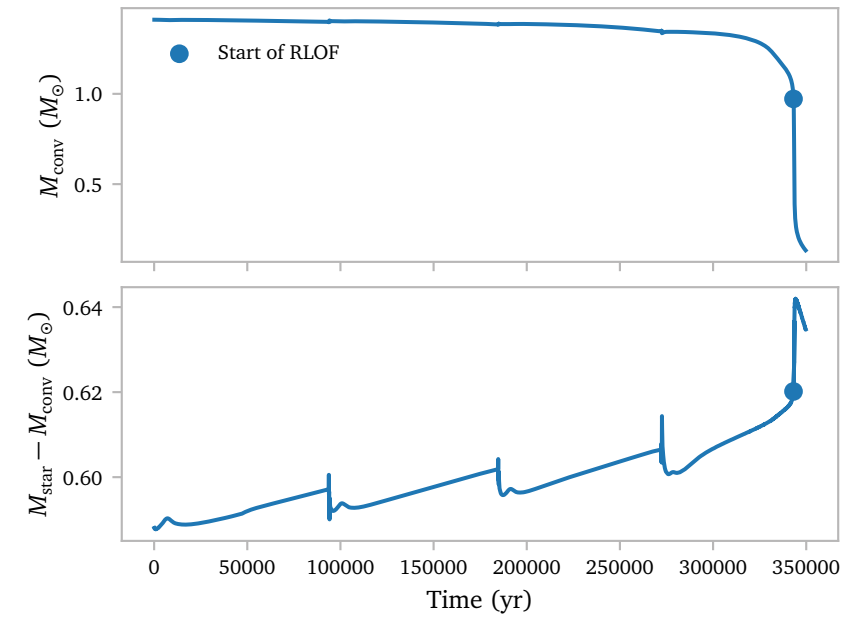
The profiles confirm that the convective envelope actually extends a lot less far to the surface as mass-transfer proceeds so the implementation for finding R_{conv} seems to be correct. I am however not sure *why* the superadiabatic region grows in radius during mass-transfer.

I also show the entropy of the profiles as a function of radius.



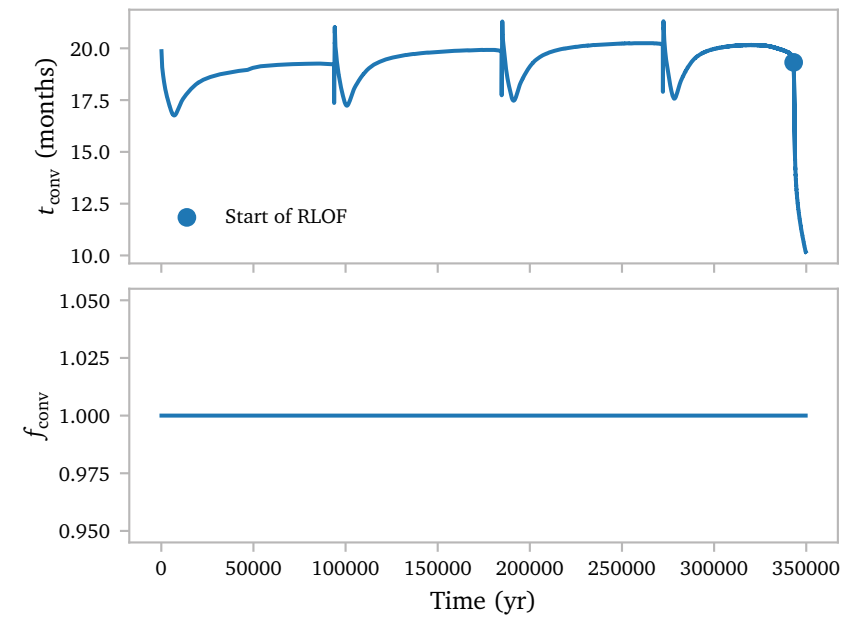
Here again, it is quite obvious that the superadiabatic region grows significant in radius.

Next I check the convective envelope mass:



I think it looks good. The mass of the convective region is roughly equal to the envelope mass, and we can even see the core mass grow between pulses. Again, the difference between the star and the convective region increases as RLOF starts, but this time, the effect is very minimal.

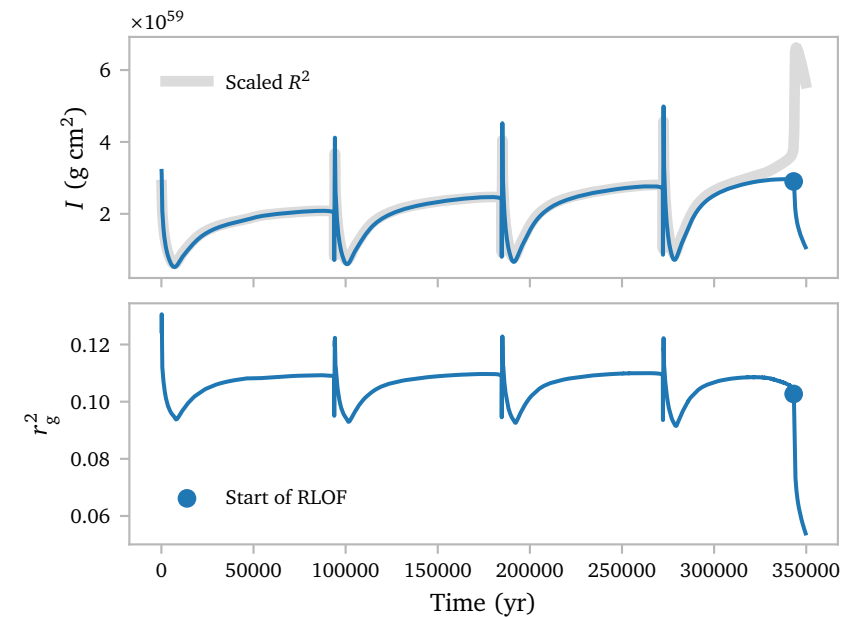
Next, I check t_{conv} and f_{conv} , where f_{conv} is a factor related to fast tides.



The convective turnover timescale seems to have realistic values. The sharp drop towards the end of the evolution is caused by the decreasing mass with similar radii and luminosities.

The fast tides factor is 1 everywhere, which is the expected result for giant stars where the tidal period is much longer than the convective turnover time.

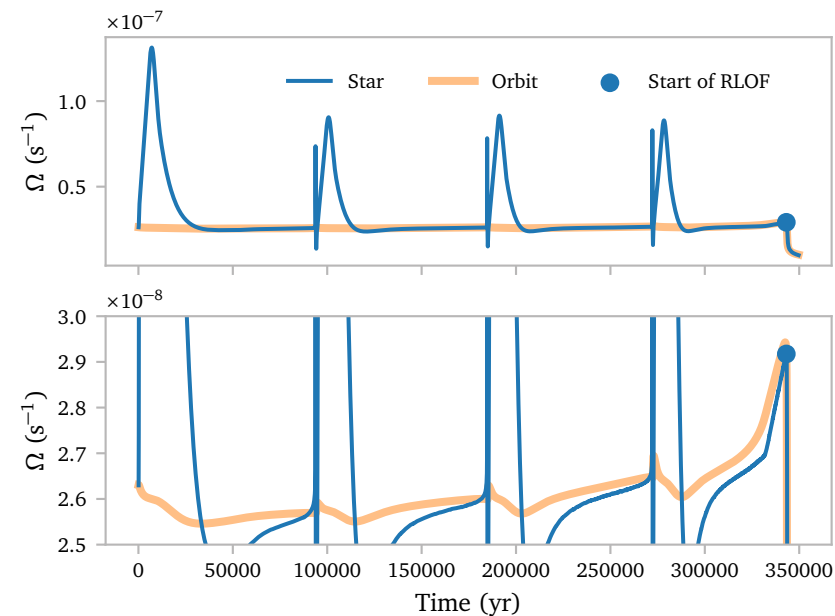
Next, the tides depend on the gyration radius r_g and the moment of inertia I . Since these are closely related, I plot them together below.



It makes sense that the moment of inertia follows the curve of R^2 really closely, since the mass barely changes. This however no longer holds once will mass loss becomes important, and especially, when RLOF starts. A quick estimation of the moment of inertia of a homogenous sphere with $M = 2M_{\odot}$ and $R = 300R_{\odot}$ gives $I \approx 7 \cdot 10^{59} \text{ g cm}^2$, which is the same order of magnitude.

The gyration radius encodes how centered the mass is concentrated in the star. For a giant star, it makes sense that $r_g^2 \approx 0.1$.

All other variables required for the [Hut \(1981\)](#) tides formalism can directly be obtained from the MESA star and binary objects. Let's look at the spin-orbit coupling.

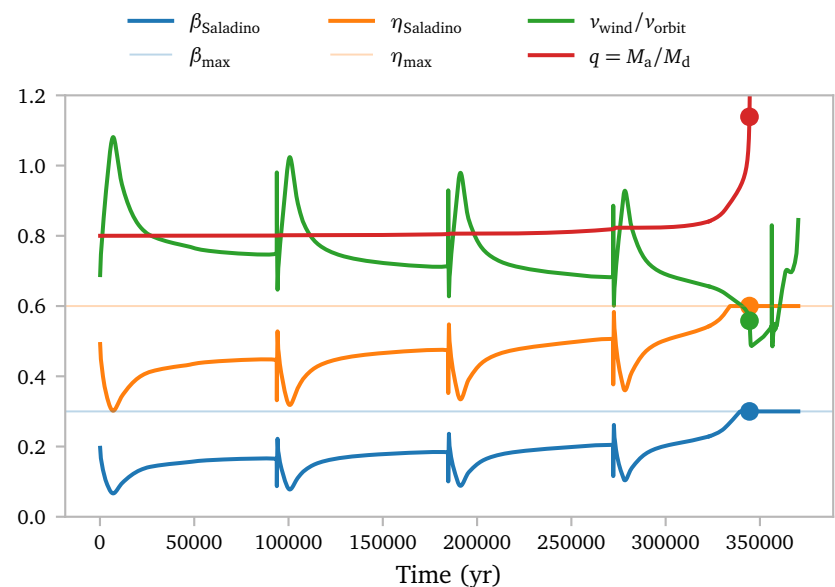


Clearly the angular frequency of the star's rotation approaches the rotation of the orbit during the interpulse period. During the thermal pulses however, the changes in moment of inertia are too fast to keep the star and orbit tidally synchronized. Interestingly as well, when RLOF starts, both the orbit and the star spin down quickly. This happens because now, both the star *and* the orbit grow significantly in size.

Without looking at the angular momentum, I think the tides look okay and seem to behave as I would expect.

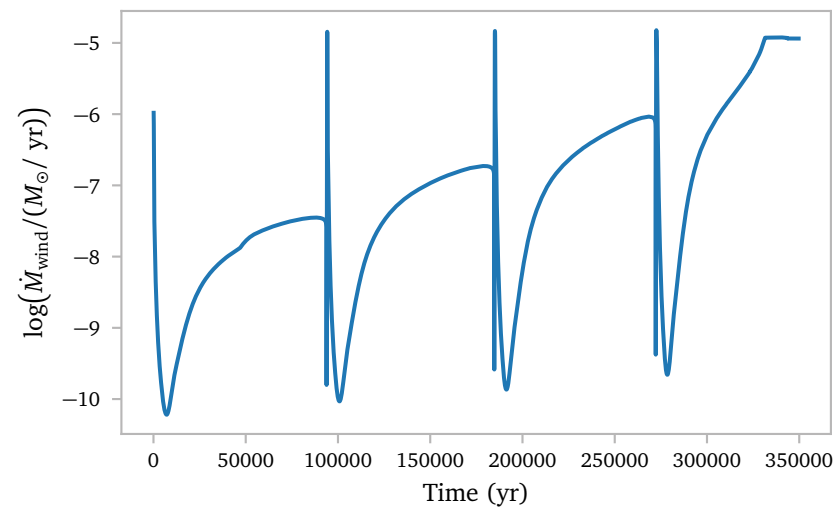
e.2 Wind

Then, let's look at the angular momentum loss due to winds, for which I follow the prescription by [Saladino et al. \(2019\)](#). The prescription depends only on the current mass loss rate due to winds and two efficiency parameters β and η , which are dependent on q and the ratio of the wind velocity to the orbital velocity $v_{\text{wind}}/v_{\text{orbit}}$.



This plot shows 4 relevant parameters for angular momentum loss due to winds. Both η and β reach their maximum value roughly at the start of significant wind mass loss while this pairs with a decrease in the ratio $v_{\text{wind}}/v_{\text{orbit}}$.

It is also interesting to look at the wind mass loss rate again, since this is a parameter in the equation for the change in angular momentum.



Interestingly, the wind mass loss depends on the stellar wind expansion velocity far from the central star v_{exp} . Now, this is obviously related to the wind velocity that I use when computing the ratio $v_{\text{wind}}/v_{\text{orbit}}$, but the prescription by [Vassiliadis & Wood \(1993\)](#) depends entirely on the pulsation period, which is dependent on the surface mass and radius, while the prescription for $v_{\text{wind}}/v_{\text{orbit}}$ depends only on the effective temperature. I want to compute both values and compare them, because I think it would be better to use a consistent wind velocity prescription.

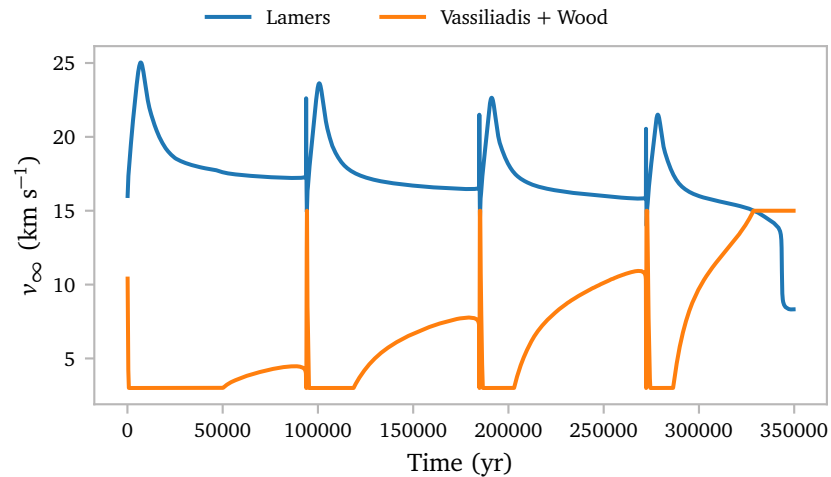
The escape velocity v_{exp} from [Vassiliadis & Wood \(1993\)](#) is defined as

$$v_{\text{exp}} (\text{km s}^{-1}) = -13.5 + 0.056P \quad \text{with} \quad \log P = -2.07 + 1.94 \log(R/R_{\odot}) - 0.9 \log(M/M_{\odot}).$$

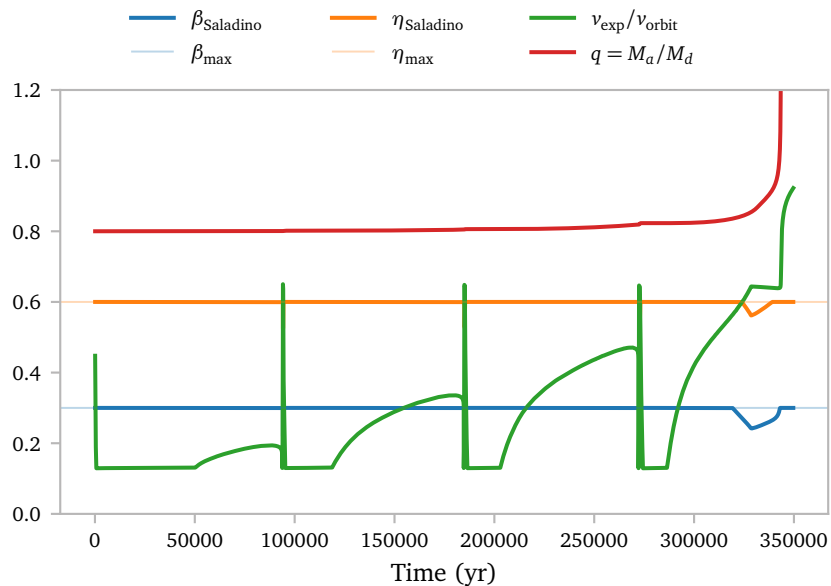
However, they limit v_{exp} between 3 and 15. We define the escape velocity from [Lamers et al. \(1995\)](#) as

$$v_{\infty} = \begin{cases} 0.35v_{\text{esc}} & \text{if } \log T_{\text{eff}} < 3.5, \\ 1.3v_{\text{esc}} & \text{if } 4.0 \leq \log(T_{\text{eff}}) < 4.35, \\ 2.6v_{\text{esc}} & \text{if } \log(T_{\text{eff}}) \geq 4.35, \\ 0.7v_{\text{esc}} + 0.7(\log T_{\text{eff}} - 4.0)v_{\text{esc}} & \text{else.} \end{cases}$$

Let me plot both for the comparison star that I have been using:

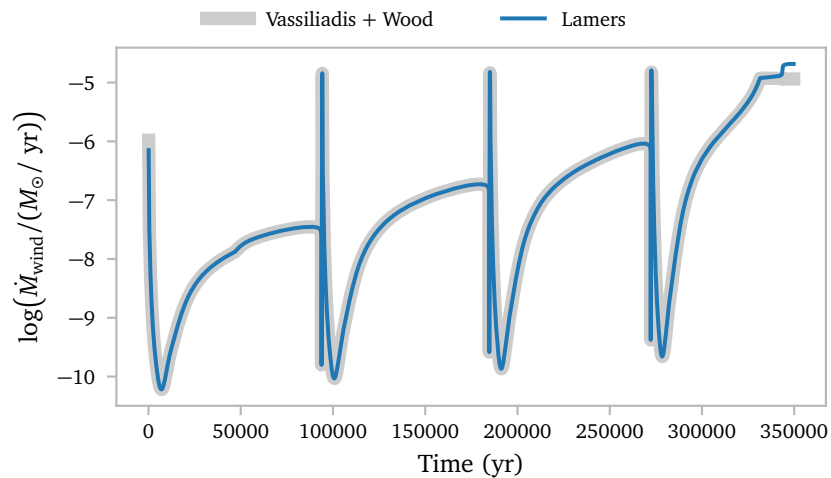


Clearly, the wind velocities are quite different. I think it is interesting to see how much this effects the values of β and η that I am using. Below, I plot β and η if I would use the wind velocity from Vassiliadis & Wood (1993).



These results make sense. Since $v_{\text{exp}} < v_{\text{wind}}$, the ratio of $v_{\text{wind}}/v_{\text{orbit}}$ will also be smaller, which in the formalism by Saladino et al. (2019) results in larger β and η . It is important to note that the simulation is being performed with the original β and η , so the results are not entirely accurate.

I a similar manner, we can compute the wind mass loss rate with the wind velocity of Lamers et al. (1995). Below, I plot this result.



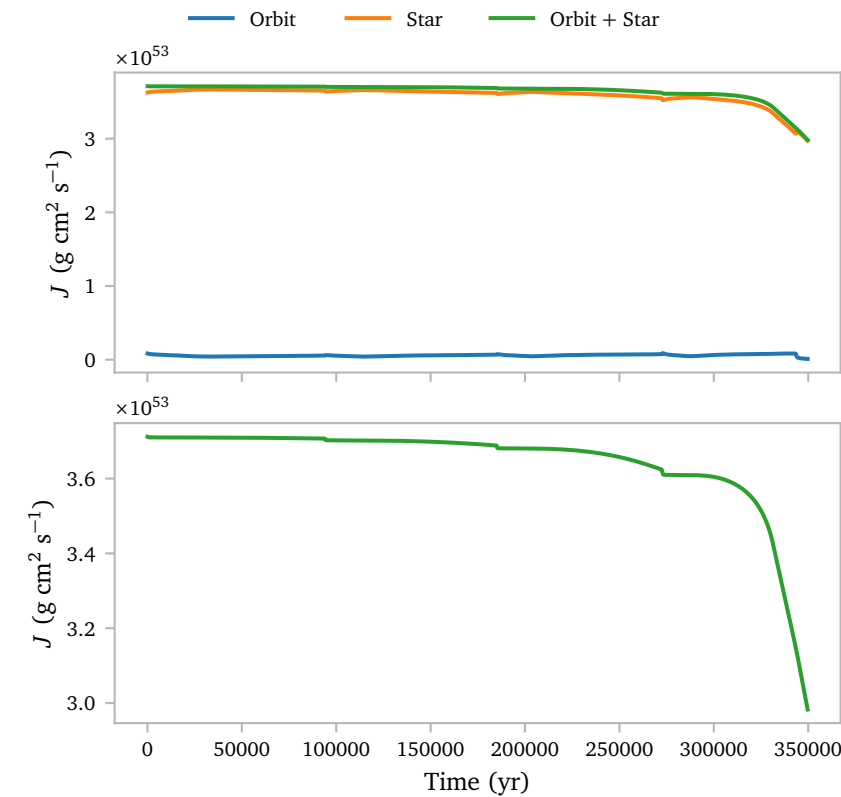
Apparently, the mass loss prescription by Vassiliadis & Wood (1993) is limited by their $\dot{M}_1$ expression for a $2M_\odot$ star, which is not dependent on the wind velocity. In this case, it really does not seem to matter which wind velocity prescription we use, expect for maybe after the mass transfer. We know however, that most of the mass is lost either before the first episode of RLOF, or during the first episode of RLOF.

I am not sure what to with this. To me, it seems wrong to use two *different* wind velocities, but I am not sure which one to choose either.

For now though, I do stick with the two prescriptions.

e.3 Angular Momentum Conservation

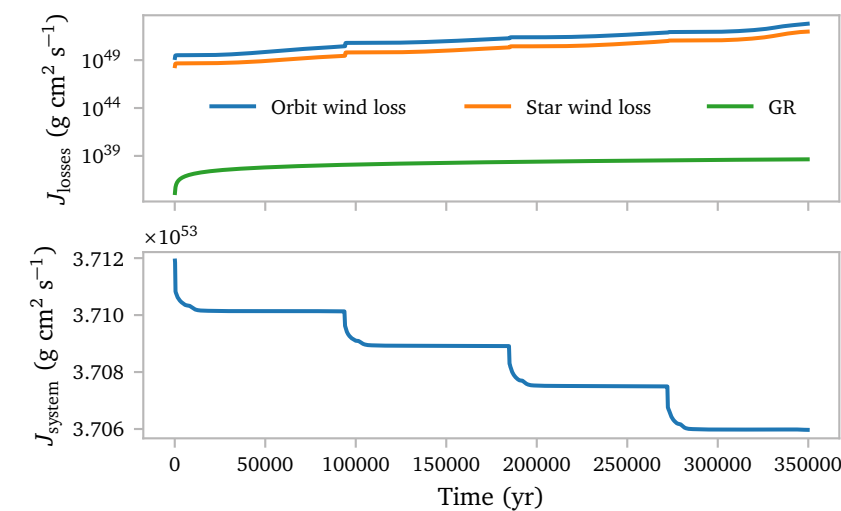
If my implementation is correct, the total angular momentum + angular momentum loss terms should be constant over time. To test this, I start by looking at the angular momentum of the star and the orbit and add them.



The top panel shows the angular momentum of the system split into the orbit and star components. As expected, the bulk of the angular momentum is stored in the orbit, while only a small portion is stored in the donor star. Since we assume mass is transferred onto a point mass, this star has *no* angular momentum. The angular momentum of the system decreases over time. This is hardly surprising; Angular momentum is lost through winds.

The bottom panel only shows the angular momentum of the system so the decreases is more clearly visible.

We can include the loss terms, after which, the total system angular momentum should be constant. By far the most important effect is the wind angular momentum loss from the orbit. Then, the angular momentum loss from wind mass loss at the surface of the star, and angular momentum loss due to gravitation waves play a minor role.



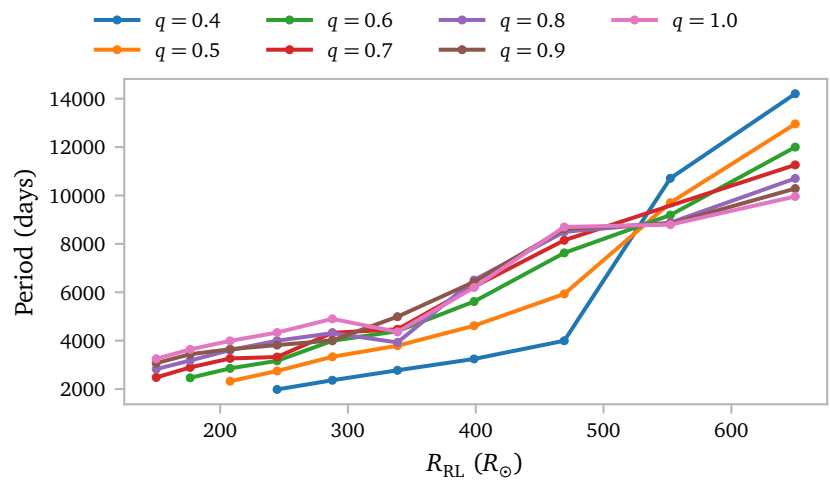
By including the losses, the system angular momentum is *almost* conserved. However, it is not *exactly* conserved and there seems to be a pattern. The drops line up perfectly with the start of the radius increase directly following the initial shrinking during a thermal pulse. Maybe this could be caused by the linear nature of the MESA code, where mass loss and angular momentum are treated with different values? Either way, I do not know for sure where the missing angular momentum is.

f. Grid

Do the results above look okay? If so, I can start a grid with these subroutines.

g. Analytical expressions of grid results

The goal of this section is to understand the results from the following plot using analytical descriptions of orbital dynamics and mass transfer:



Let's start with the general expression for angular momentum of a two body problem of mass masses in the centre of mass coordinate system:

$$\vec{J} = \mu \vec{r} \times \vec{v} = \mu r^2 \dot{\phi} \, \hat{r} \times \hat{\phi}.$$

Here, μ is the reduced mass defined as

$$\mu = \frac{M_1 M_2}{M_1 + M_2}.$$

The magnitude of the orbital angular momentum is then

$$J = \mu r^2 \dot{\phi} = \text{const.}$$

Energy is a conserved quantity as well, which we can express as

$$\begin{aligned} E &= K + V = \frac{1}{2} \mu \dot{r}^2 - G \frac{M_1 M_2}{r} \\ &= \frac{1}{2} \mu \dot{r}^2 + \frac{1}{2} \mu r^2 \dot{\phi}^2 - G \frac{M_1 M_2}{r}. \end{aligned}$$

By substituting the equation for the angular momentum, we find that

$$E = \frac{1}{2} \mu \dot{r}^2 + \frac{J^2}{2 \mu r^2} - G \frac{M_1 M_2}{r}.$$

Combining this with the standard result for the specific energy of an elliptical orbit,

$$\varepsilon = -G \frac{M_1 + M_2}{2a} \implies E = \varepsilon \mu = -G \frac{M_1 M_2}{2a},$$

we can solve for J^2 :

$$\begin{aligned} -G \frac{M_1 M_2}{2a} &= \frac{1}{2} \mu \dot{r}^2 + \frac{J^2}{2 \mu r^2}, \\ \frac{J^2}{2 \mu r^2} &= G M_1 M_2 \left[\frac{1}{r} - \frac{1}{2a} \right] - \frac{1}{2} \mu \dot{r}^2. \\ J^2 &= 2 G r \frac{M_1^2 M_2^2}{M_1 + M_2} \left[1 - \frac{r}{2a} \right] - \mu^2 r^2 \dot{r}^2. \end{aligned}$$

Using that total energy is conserved throughout the orbit, we can solve for J^2 at one point in the orbit, and the result will hold throughout the orbit. At the periapsis or apoasis, $\dot{r} = 0$, which reduces the equation to

$$J^2 = 2 G r_p \frac{M_1^2 M_2^2}{M_1 + M_2} \left[1 - \frac{r_p}{2a} \right].$$

Using $r_p = a(1 - e)$, we find that

$$\begin{aligned} J^2 &= 2 G \frac{M_1^2 M_2^2}{M_1 + M_2} \left[\frac{1 + e}{2} \right] a (1 - e). \\ J^2 &= G \frac{M_1^2 M_2^2}{M_1 + M_2} a (1 - e^2). \end{aligned}$$

These are all terms we *know* so we can use this definition of J as a conserved quantity and explore what happens when we vary masses due to mass transfer. It is important to note that this definition assumes the angular momentum of the binary is strictly the orbital angular momentum of the stars. For our model without tides (and thus no stellar rotation), this is perfectly fine but the upcoming results will not strictly be true for rotating stars.

Let's start by taking the time derivative of J^2 :

$$\frac{d}{dt} = 2J \frac{dJ}{dt} = \frac{d}{dt} \left[G \frac{M_1^2 M_2^2}{M_1 + M_2} a (1 - e^2) \right].$$

It is actually easier to compute $\frac{d}{dt} (J^2) / J^2$, since a lot of the terms cancel.

$$\begin{aligned} \frac{2}{J} \frac{dJ}{dt} &= \frac{1}{J^2} \frac{d}{dt} \left[G \frac{M_1^2 M_2^2}{M_1 + M_2} a (1 - e^2) \right], \\ \frac{2}{J} \frac{dJ}{dt} &= \frac{1}{J^2} \left\{ \frac{d}{dt} [M_1^2] \frac{J^2}{M_1^2} + \frac{d}{dt} [M_2^2] \frac{J^2}{M_2^2} + \frac{d}{dt} [(M_1 + M_2)^{-1}] \cdot \right. \\ &\quad \left. J^2 (M_1 + M_2) + \frac{d}{dt} [a] \frac{J^2}{a} + \frac{d}{dt} [1 - e^2] \frac{J^2}{1 - e^2} \right\}, \\ \frac{2}{J} \frac{dJ}{dt} &= \frac{2 \dot{M}_1}{M_1} + \frac{2 \dot{M}_2}{M_2} - \frac{\dot{M}_1 + \dot{M}_2}{M_1 + M_2} + \frac{\dot{a}}{a} - \frac{2e \dot{e}}{1 - e^2}. \end{aligned}$$

In the case of conservative mass transfer in a circular orbit and no angular momentum loss due to wind mass loss, $\dot{J} = 0$, $e = 0$ and $\dot{M}_1 = -\dot{M}_2$. This allows us to solve for the change in separation, which is closely related to the period.

$$\frac{\dot{a}}{a} = \frac{2 \dot{M}_d}{M_a} - \frac{2 \dot{M}_d}{M_d} = 2 \frac{\dot{M}_d}{M_d} \left(\frac{M_d}{M_a} - 1 \right).$$

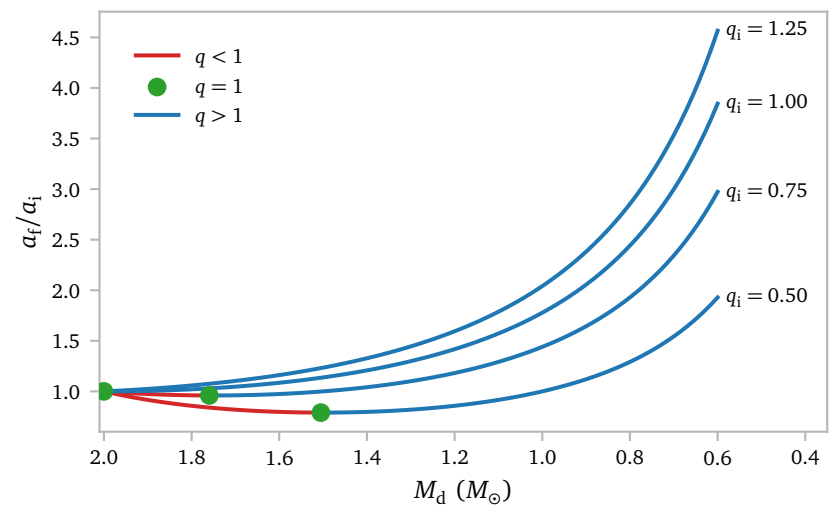
Since mass is conserved, $M_a = M_{\text{tot}} - M_d$ is always true and M_{tot} is constant. This means we can write

$$\begin{aligned} \frac{\dot{a}}{a} &= \frac{2 \dot{M}_d}{M_d} \left(\frac{M_d}{M_{\text{tot}} - M_d} - 1 \right), \\ &= 2 \frac{\dot{M}_d}{M_d} \left(\frac{2 M_d - M_{\text{tot}}}{M_{\text{tot}} - M_d} \right), \\ \frac{da}{a} &= \frac{2}{M_d} \left(\frac{2 M_d - M_{\text{tot}}}{M_{\text{tot}} - M_d} \right) dM_d, \\ \int_{a_i}^{a_f} \frac{da}{a} &= \int_{M_{d,i}}^{M_{d,f}} \frac{2}{M_d} \left(\frac{2 M_d - M_{\text{tot}}}{M_{\text{tot}} - M_d} \right) dM_d, \\ \ln \left(\frac{a_f}{a_i} \right) &= 2 \int_{M_{d,i}}^{M_{d,f}} \frac{2 M_d - M_{\text{tot}}}{M_d (M_{\text{tot}} - M_d)} dM_d, \end{aligned}$$

Splitting the fraction yields

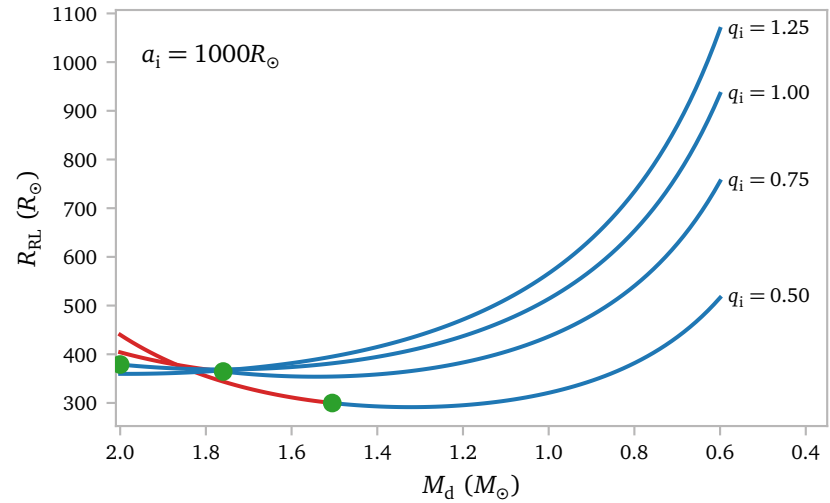
$$\begin{aligned} \ln \left(\frac{a_f}{a_i} \right) &= 2 \int_{M_{d,i}}^{M_{d,f}} -\frac{1}{M_d} + \frac{1}{M_{\text{tot}} - M_d} dM_d, \\ \ln \left(\frac{a_f}{a_i} \right) &= 2 \ln \left(\frac{M_{d,i}}{M_{d,f}} \right) + 2 \ln \left(\frac{M_{\text{tot}} - M_{d,i}}{M_{\text{tot}} - M_{d,f}} \right) \\ \frac{a_f}{a_i} &= \left(\frac{M_{d,i} (M_{\text{tot}} - M_{d,i})}{M_{d,f} (M_{\text{tot}} - M_{d,f})} \right)^2. \end{aligned}$$

Cool, Let's show this relation in a couple of different ways starting with an intuitive one, the ratio between the initial and final orbit as a function of the donor mass.



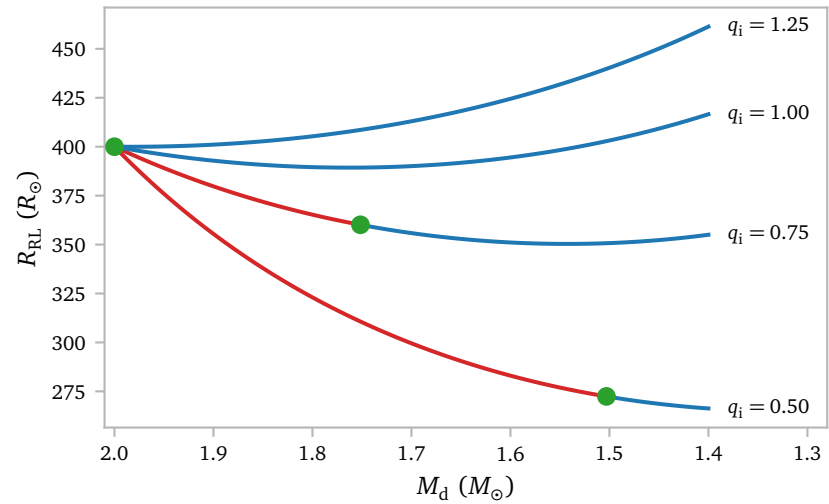
The minimum in separation happens nicely at the point where $q = 1$, exactly as 1 would expect.

Next, we can plot the same thing, but now as a function of the Roche lobe radius.

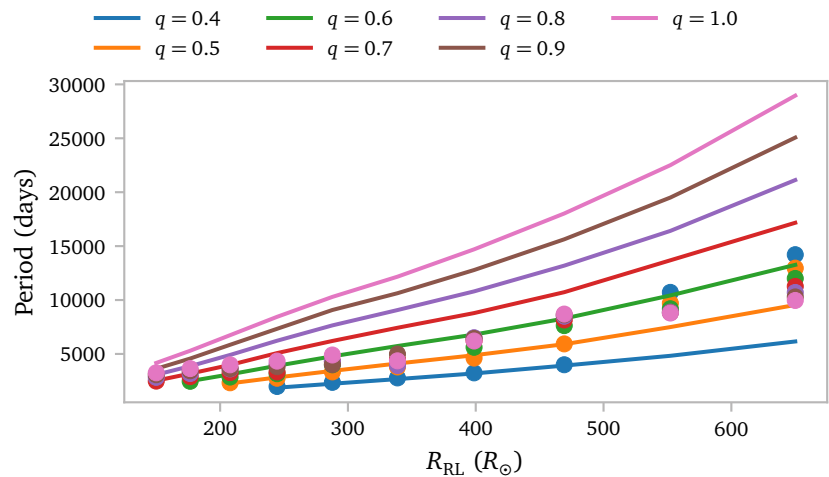


When evaluating Roche Lobe radii instead of separations, the minimum does not happen exactly when $q = 1$.

We can also start the evolution at a specific Roche lobe radius, similar to what we do in the MESA models:



If we apply this model of fully conservative mass transfer⁵ to the initial and final conditions of the MESA models, we get the following results:



5 With the assumption of *no* wind mass loss.

Except for maybe the very low mass ratios, the fully conservative model overestimates the periods a lot.

It is important to note that the mass transfer efficiency is actually not 1 for these models due to the radiation corrected mass transfer rate.

We *could* actually just numerically solve the full $\dot{J}/J$ differential equation:

$$\frac{2}{J} \frac{dJ}{dt} = \frac{2\dot{M}_1}{M_1} + \frac{2\dot{M}_2}{M_2} - \frac{\dot{M}_1 + \dot{M}_2}{M_1 + M_2} + \frac{\dot{a}}{a} - \frac{2e\dot{e}}{1-e^2}.$$

Rearranging assuming $e = 0$ gives

$$\frac{\dot{a}}{a} = \frac{2\dot{J}}{J} - \frac{2\dot{M}_1}{M_1} - \frac{2\dot{M}_2}{M_2} + \frac{\dot{M}_1 + \dot{M}_2}{M_1 + M_2}.$$

If we have fine samples of these terms, the orbital evolution comes out of this equation quite naturally. Below I show a simple piece of python code that numerically solves for the final separation:

```
def integrate_a(a_i, Md, Ma, Mdot_d, Mdot_a, t, Jdot_over_J):
    ln_a = np.log(a_i)

    for i in range(len(t) - 1):
        dt = t[i + 1] - t[i]

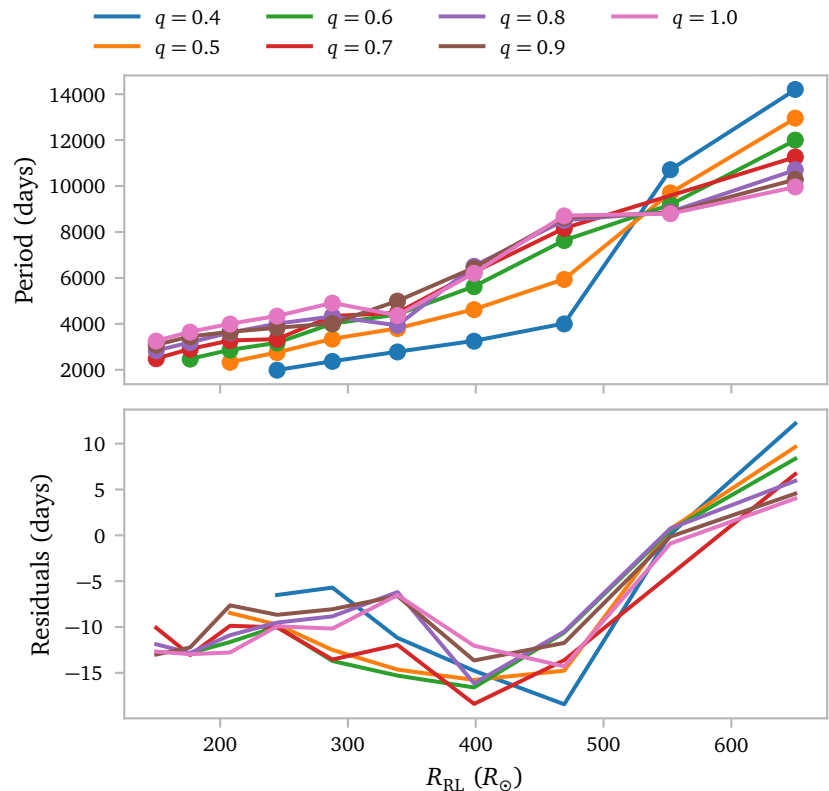
        Md_mid = 0.5 * (Md[i] + Md[i + 1])
        Ma_mid = 0.5 * (Ma[i] + Ma[i + 1])
        Ma_mid = 0.5 * (Ma[i] + Ma[i + 1])
        Mtot_mid = Md_mid + Ma_mid
        Mdot_d_mid = 0.5 * (Mdot_d[i] + Mdot_d[i + 1])
        Mdot_a_mid = 0.5 * (Mdot_a[i] + Mdot_a[i + 1])
        Jdot_over_J_mid = 0.5 * (Jdot_over_J[i] + Jdot_over_J[i+1])

        dlnadt = (
            2 * Jdot_over_J_mid
            - 2 * Mdot_d_mid / Md_mid
            - 2 * Mdot_a_mid / Ma_mid
            + (Mdot_d_mid + Mdot_a_mid) / Mtot_mid
        )

        ln_a += dlnadt * dt

    return np.exp(ln_a)
```

The MESA history files actually contain *all* of the required values to compute the expected final separation. Below I plot them and the actual results from MESA.



The *dots* are the MESA values, the lines are the integrated values using the differential equation.

Here, I plot the residuals. They are very small and well within the expected error, especially taking into account that the history data for these models only contains data for 1 in every 10 model steps.

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